

Learning Objective 1.3: I can determine the polarization of an antenna and describe the bandwidth characteristics of common antenna types.

Show your work. A **Documentation** line at the end is required (write **None** if you did not collaborate).

1. **Identify the polarization.** For each plane wave propagating in $+\hat{\mathbf{z}}$, identify the polarization (linear, RHCP, LHCP, or elliptical) and — where linear — the tilt angle relative to $\hat{\mathbf{x}}$.

(a) $\vec{E} = \hat{\mathbf{x}}3\cos(\omega t - kz)$

(b) $\vec{E} = \hat{\mathbf{x}}2\cos(\omega t - kz) + \hat{\mathbf{y}}2\cos(\omega t - kz)$

(c) $\vec{E} = \hat{\mathbf{x}}5\cos(\omega t - kz) + \hat{\mathbf{y}}5\cos(\omega t - kz - 90^\circ)$

(d) $\vec{E} = \hat{\mathbf{x}}5\cos(\omega t - kz) + \hat{\mathbf{y}}5\cos(\omega t - kz + 90^\circ)$

(e) $\vec{E} = \hat{\mathbf{x}}3\cos(\omega t - kz) + \hat{\mathbf{y}}1\cos(\omega t - kz - 90^\circ)$

2. **Axial ratio.** A satellite downlink antenna is spec'd circularly polarized with $\text{AR} \leq 3 \text{ dB}$ across its operating band.

- (a) Convert the 3 dB axial ratio to a linear ratio (major/minor axis).

- (b) Write the two orthogonal linear components (up to a common scale) with the appropriate 90° phase, assuming the ellipse major axis is along $\hat{\mathbf{x}}$.

- (c) An $\hat{\mathbf{x}}$ receiver picks off E_x ; a $\hat{\mathbf{y}}$ receiver picks off E_y . Compute the ratio of received *power* in dB. Should it match part (a)?

3. Polarization loss factor.

- (a) A ground station transmits a *linearly* polarized signal along $\hat{\mathbf{x}}$. A satellite receiver is **RHCP** and pointed correctly. Compute the PLF in dB.
- (b) Same ground station, but the satellite receiver is **LHCP**. Compute the PLF.
- (c) Two linear whip antennas: radio A vertical, radio B tilted 30° from vertical. Compute the PLF.
- (d) In one sentence, explain why aircraft-to-ground links almost always specify circular polarization on at least one end.

4. **Impedance bandwidth from a VSWR spec.** An antenna is spec'd “VSWR ≤ 2 from 2.30GHz to 2.55GHz.”

(a) What is the impedance bandwidth in MHz?

(b) What is the fractional bandwidth (FBW) in percent?

(c) Convert VSWR = 2 to $|\Gamma|$ and to return loss in dB.

(d) Narrowband, broadband, or ultra-wideband? Likely family — patch, horn, or log-periodic?

5. Fractional and ratio bandwidth.

- A cellular antenna covers 698–960MHz (low) and 1.71–2.69GHz (high). Compute the FBW and RBW of each band.
- A UWB radar antenna covers 3.1–10.6GHz. Compute its RBW. Does it meet the FCC UWB threshold of $\text{RBW} \geqslant 2\text{:}1$?
- Which antenna family would you pick for each of the three bands, with a one-sentence reason?

6. **Chu-Harrington intuition.** A design must fit inside a sphere of radius $a = 15$ mm and operate at $f = 915$ MHz (ISM band).

(a) Compute λ , k , and the electrical size ka .

(b) Estimate the minimum Q from $Q_{\min} \approx \frac{1}{(ka)^3} + \frac{1}{ka}$.

(c) Estimate the maximum fractional bandwidth $\text{FBW}_{\max} \approx 1/Q_{\min}$ (for $\text{VSWR} \leq 2$).

(d) The designer wants 100 MHz of usable bandwidth at 915 MHz. Possible in a 15 mm sphere?

7. **Mismatch that kills the amplifier.** A power amplifier (PA) delivers a forward power of $P_{\text{fwd}} = +40$ dBm toward a transmit antenna. A shunt protection diode at the PA output fails catastrophically if it must dissipate more than $P_{\text{max}} = +30$ dBm. Model *all* of the reflected power as landing on the diode.

(a) Convert P_{fwd} and P_{max} to watts.

(b) In terms of $|\Gamma|$ at the antenna, how much power is reflected back toward the PA?

(c) Find the largest $|\Gamma|$ the system can tolerate before the diode is destroyed.

(d) Convert to a VSWR limit and a return-loss limit, and state the datasheet rule.

(e) In one sentence: how does a circulator or isolator between the PA and antenna change this picture?

Documentation: _____